

Zoe Hay

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Work Experience

Software Engineer, Contract @ Lavender

Summer 2023

- Built an automated E2E test suite in **TypeScript** using **Selenium WebDriver** for Lavender's core product, a Chrome extension generative-AI powered email assistant and an admin web application
- Authored automated tests for core user flows such as upgrading billing plans and using features from the extension's toolbar, which enabled smooth launch of a new billing system

Senior Engineer, Design Evaluation Engineering @ Analog Devices

2018 - 2022

- Developed an automated test suite in **Python** to set up test conditions by controlling equipment, such as power supplies and a microcontroller, then collect and process measurements to automatically evaluate integrated circuit products reducing evaluation time from days to hours
- Supported development of new power product integrated circuits, detected and resolved issues, performed failure analysis
- Collected datasheet characterization curve measurements, coordinated and debugged with test engineers
- Designed and built boards for lab evaluation (schematic drawing, component selection, assembly)

Design Engineer Intern @ Linear Technology, Analog Devices

Summer 2017

- Investigated dielectric characteristics for use in high performance linear regulator applications, characterized the effect of set capacitor on low dropout regulator performance, performed technical investigations for the LDO team
- Reworked demo circuits for novel applications, integrated off-the-shelf components into real-world applications

Electrical Engineering Intern @ Rantec Power Systems

Summer 2016

- Assisted electrical engineering team in the design and manufacturing of power supplies and power conversion products
- Updated assembly procedures and testing documentation, investigated and resolved manufacturing issues
- Simulated and analyzed circuits, sized components based on system requirements

Projects

Personal Website www.zoemhay.com

2023 - Ongoing

Provisioned and managed **AWS** infrastructure (EC2, VPC, EIP, security groups) with **Terraform**. Containerized applications with **Docker Compose** behind an **NGINX** reverse proxy with TLS via Let's Encrypt/Certbot. Shell scripts to automate instance initialization, service setup via systemd, and deployment workflows.

Guild Wars 2 Inventory www.gw2-armory.com

2024 - Ongoing

Allows players to view inventories across all characters using the public GW2 API. Backend: **Go**, **Gin**, **GORM**, **PostgreSQL**. Frontend: **React**, **TypeScript**.

E-Commerce Application e-commerce.zoemhay.com

2022

An e-commerce web application that supports a shopping cart for authenticated users and a separate admin interface for managing products and users. Backend: **ExpressJS**, **Prisma ORM**, **PostgreSQL**, **Jest**. Frontend: **React**, **React Router**, **Styled Components**. Functional testing with **Playwright**.

Angular E-Commerce Application angular-e-commerce.zoemhay.com

2023

A rewrite of E-Commerce Application's frontend using **Angular** and **TypeScript**.

Publications

Kappa Switching DC-DC Converter with Continuous Input and Output Currents

Summer 2019

Physically implemented and performed testing for a new converter topology invented by colleagues. Collected waveform captures, efficiency and EMI measurements.

Multiphase Implementation of Modified Boost Converter

Fall 2017 - Spring 2018

Designed and implemented a 95W, two-phase interleaved, modified boost converter with reduced input current ripple. Performed performance comparison with a standard boost converter.

DC House Energy Management System

Fall 2016 - Spring 2017

Implemented and improved an energy management system adding a bidirectional DC-DC converter. Designed to integrate into the DC House project.

Education and About

California Polytechnic State University, San Luis Obispo

B.S. + M.S. Electrical Engineering, June 2018

Electrical Engineer by formal education. Software Engineer by practice.